Objective Weighting Methods

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Introduction - IDOCRIW

The Integrated Determination of Objective Criteria Weights(IDOCRIW) method was introuduced in 2016. This technique benefits from the Entropy and Criterion Impact LOSs methods to determine a relative impact loss as well as the weight of attributes in a combination with two methods.

This methods has the following features:

- It is one of the compensatory methods;
- Attributes are independent;
- The qualitative attributes should be converted into the quantitative attributes.

The Decision Matrix

$$X = \begin{bmatrix} r_{11} & \dots & r_{1j} & \dots & r_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ r_{i1} & \dots & r_{ij} & \dots & r_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ r_{n1} & \dots & r_{nj} & \dots & r_{nn} \end{bmatrix}_{n \times n}$$

In this decision matrix, r_{ij} is the element of the decision matrix for ith alternative in jth attribute.

Description of IDOCRIW Method

- First, obtain the entropy weight.

The Degree of Entropy

$$E_j = -\frac{1}{\ln n}\sum_{i=1}^{n} r_{ij} \cdot \ln r_{ij}; \quad j = 1, \cdots, n, \quad 0 \le E_j \le 1 \tag{1}$$

The Entropy Weight

$$w_j = \frac{1 - E_j}{\sum_{j=1}^{n}(1 - E_j)} \tag{2}$$

Description of IDOCRIW Method

- Then, obtain the criterion impact loss weight.

Determine a square matrix

$$a_j = \max_i r_{ij} = a_{k_i j}; \quad i, j \in \{1, \cdots, n\} \tag{3}$$

$a_{k_i j}$ specifies the maximum values of the jth criteria, and the ith row of the square matrix contains the elements of the row k_i of decision matrix.
In the square matrix, $a_{ij} = a_{k_i j}$ and $a_{jj} = a_j$.

The Relative Impact Loss

$$p_{ij} = \frac{a_{jj} - a_{ij}}{a_{jj}} \tag{4}$$

p_{ij} represents the impact loss of the jth attribute, if selected as the best value.

Description of IDOCRIW Method

The Weight System Matrix

Given the values of p_{ij}, the weight system matrix can be formed.

$$F = \begin{bmatrix} -\sum_{i=1}^{n} p_{i1} & p_{12} & \cdots & p_{1n} \\ p_{21} & -\sum_{i=1}^{n} p_{i2} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & -\sum_{i=1}^{n} p_{in} \end{bmatrix}_{n\times n}$$

The Criteria Impact Loss Weight

The weight of the attributes $\begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}$ is determined through the solotion of the following equation.

$$Fq^T = 0 \tag{5}$$

Description of IDOCRIW Method

- Finally, we arrive at the aggregate weight.

The Aggregate Weight

Considering the entropy weight and CILOS weight, the aggregate weight values of the attributes is determined by the following equation.

$$\omega_j = \frac{q_j \cdot w_j}{\sum_{j=1}^{n} q_j \cdot w_j} \tag{6}$$

Introduction - CRITIC

The Criteria Importance Through Intercriteria Correlation (CRITIC) method was proposed in 1995.

This methods has the following features:

- No need for the independence of attributes;
- The qualitative attributes should be converted into the quantitative attributes.

The Decision Matrix

$$X = \begin{bmatrix} r_{11} & \cdots & r_{1j} & \cdots & r_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ r_{i1} & \cdots & r_{ij} & \cdots & r_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ r_{m1} & \cdots & r_{mj} & \cdots & r_{mn} \end{bmatrix}_{m \times n}$$

In this decision matrix, r_{ij} is the element of the decision matrix for ith alternative in jth attribute.

Description of Critic Method

The Correlation Coefficient

The correlation coefficient among attributes is determined by the following equation. ρ_{jk} is the correlation coefficient between jth and kth attributes.

$$\rho_{jk} = \frac{\sum_{i=1}^{m}(x_{ij} - \overline{x_j})(x_{ik} - \overline{x_k})}{\sqrt{\sum_{i=1}^{m}(x_{ij} - \overline{x_j})^2 \sum_{i=1}^{m}(x_{ik} - \overline{x_k})^2}} \tag{7}$$

where $\overline{x_j}$ and $\overline{x_k}$ display the mean of jth and kth attributes.

The Index C

First, obtain the standard deviation of each attribute.

$$\sigma_j = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(x_{ij} - \overline{x_j})^2}; \quad i = 1, \cdots, m \tag{8}$$

Description of Critic Method

The Index C

Then, calculate the index C using the following equation.

$$C_j = \sigma_j \sum_{k=1}^{n} (1 - \rho_{jk}); \quad j = 1, \cdots, n \tag{9}$$

- Finally, we have the weight of attributes.

The Weight of Attributes

$$w_j = \frac{C_j}{\sum_{j=1}^{n} C_j}; \quad j = 1, \cdots, n \tag{10}$$

The End